

Reinforcement Learning

What Is Reinforcement Learning?

- A way for computers to learn by trial and error.
- An agent interacts with an environment to achieve a goal.
- The agent learns which actions lead to the best results.

Key Ideas

- Agent: the learner or decision maker.
- Environment: the world the agent interacts with.
- State: the current situation.
- Action: what the agent can do.
- Reward: feedback after each action.

The Goal

- The agent wants to maximize total reward over time.
- It learns a strategy, called a policy, to decide the best action for each situation.

How Learning Happens

- 1. The agent takes an action.
- 2. The environment gives a new state and a reward.
- 3. The agent updates what it has learned.
- 4. Repeat until behavior improves.

Introducing Q-Learning

- Q-Learning helps the agent learn the value of each action in each state.
- 'Q' means quality — how good an action is in a specific situation.
- It stores these values in a table called the Q-table.

Q-Learning Example

- Given a 4x4 grid, with current state at (0,0)
- Objective: Move to (3,3)
- Possible actions: Left, Right, Up, Down
- Reward (R):
 - Each movement: -1
 - For reaching (3,3): 10
- Learning Rate (LR): 0.5
 - How fast the agent learns
- Discount Factor (DF) : 0.9
 - How much emphasis on future rewards

Q-Learning Example

State	Left	Right	Up	Down
(0,0)	-1	-0.3	-1	-0.5
(0,1)	0.0	0.4	0.2	0.1

- Take right action, highest reward of -0.3
- Update table:
 - $Q = Q_{\text{old}} + LR * (R + DF * \max Q(0,1) - (-0.3))$
 - $= -0.3 + 0.5 * (-1 + 0.9 * 0.4 + 0.3)$
 - $= -0.47$

Q-Learning Example

State	Left	Right	Up	Down
(0,0)	-1	-0.3 -> -0.47	-1	-0.5
(0,1)	0.0	0.4	0.2	0.1

- Q value for State (0,0) Down action:
 - Updated from -0.3 to -0.47

How the Q-Table Updates

- When the agent takes an action and gets a reward:
 - - It updates the Q-value for that state-action pair.
 - - Over time, the table gets more accurate.
 - - The agent starts choosing the best actions more often.

Exploration vs Exploitation

- Exploration: try new actions to discover rewards.
- Exploitation: choose the best-known action.
- A good balance between both helps learning.

Summary

- Reinforcement learning = learning from experience.
- Q-learning = a simple way to store and improve action values.
- Over time, the agent learns what works best to get the most reward.